Ayush Lahiri

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Computer Science & Engineering student at IIIT Kalyani and App Dev Lead at GDG IIIT Kalyani, working at the intersection of ML research, systems programming, and product engineering. Experienced in LLM fine-tuning (LoRA/QLoRA on LLaMA), custom SVM kernel design, and GPU kernel programming with OpenAI Triton. Proficient in Python (PyTorch, HuggingFace, scikit-learn), Android development, with a strong foundation in data structures, algorithms, and OS internals.

Education

- **Bachelor's of Technology in Computer Science** ***June 2029(Exp.)***

  Indian Institute of Information Technology Kalyani, India

- **High School Diploma, CBSE** ***April 2025***

  Mothers' International School

- **Secondary School Education, ICSE** ***April 2023***

  Carmel School, Madhupur

- **CS229: Machine Learning** ***2026***

  Stanford University (Online) — Supervised/unsupervised learning, SVMs, neural networks, regularization

- **Hands-on Deep Learning** ***2025***

  MIT OpenCourseWare — CNNs, RNNs, Transformers, attention mechanisms, practical deep learning pipelines

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Skills

- Android development
- Web Development
- Ruby on Rails
- Mobile app creation
- Native Android Optimization
- Data structures and algorithms

- Application security
- Machine Learning (PyTorch, TensorFlow, scikit-learn)
- Deep Learning & LLM Fine-Tuning (LoRA, QLoRA, HuggingFace PEFT)
- GPU Kernel Programming (OpenAI Triton)
- Python (NumPy, Pandas, Matplotlib, bitsandbytes)
- Kernel Methods & SVMs

Experience

Machine Learning & Deep Learning Projects

- **LLaMA Fine-Tuning via LoRA & QLoRA:** Parameter-efficient fine-tuning of LLaMA using Low-Rank Adaptation and 4-bit NF4 quantization (QLoRA via bitsandbytes), enabling single-GPU training of 2B+ parameter models with minimal quality loss. Leveraged HuggingFace PEFT and Transformers.
- **Custom SVM Kernels:** Designed non-standard positive-definite kernel functions for SVMs to handle structured and graph-based feature spaces where standard RBF and polynomial kernels fail. Validated using kernel alignment metrics with scikit-learn.
- **Triton GPU Kernel Programming:** Wrote fused GPU kernels in OpenAI Triton, bypassing PyTorch dispatch overhead with hand-controlled tile sizes, vectorized loads, shared memory staging, and warp-level reductions. Targeted bottleneck operations in attention and linear layers.
- **Leaf-Health (Image Classification):** Built a deep learning model to classify leaf images as healthy or blister-blight affected. Applied CNN-based architectures for binary image classification with practical preprocessing and training pipelines.

Work Experience

App Dev Lead

GDG IIIT Kalyani, Kalyani, India *October 2025-Current*

- Designed user interfaces that engaged multiple senses and produced immersive experiences.
- Maintained comprehensive knowledge of mobile development cycle and addressed challenges arising in each phase.
- Coded Website using HTML, CSS, JAVASCRIPT AND PYTHON(FLASK) languages